

Introduction

Genetic testing with whole-exome sequencing and whole-genome sequencing is a powerful way to quickly identify the pathogenic variants for roughly 40% of all patients with suspected genetic disorders. For the unsolved cases, variants of uncertain significance (VUSs) are frequently reported, indicating their uncertain associations with the disease phenotypes due to limited knowledge at the time of diagnosis. Frequent reanalysis is needed to reassess these VUS's as new knowledge is gained. If done manually, this presents significant challenges.

Here, we introduce an AI-based Retrieval-Augmented Generation (RAG) system designed to address these challenges by automating the reclassification of VUS and providing clinicians with up-to-date, evidence-based genomic insights.

Workflow

1. PDF Upload & Processing

- Upload clinical genetic reports through the web interface
- Identify variants, genes, and clinical classifications

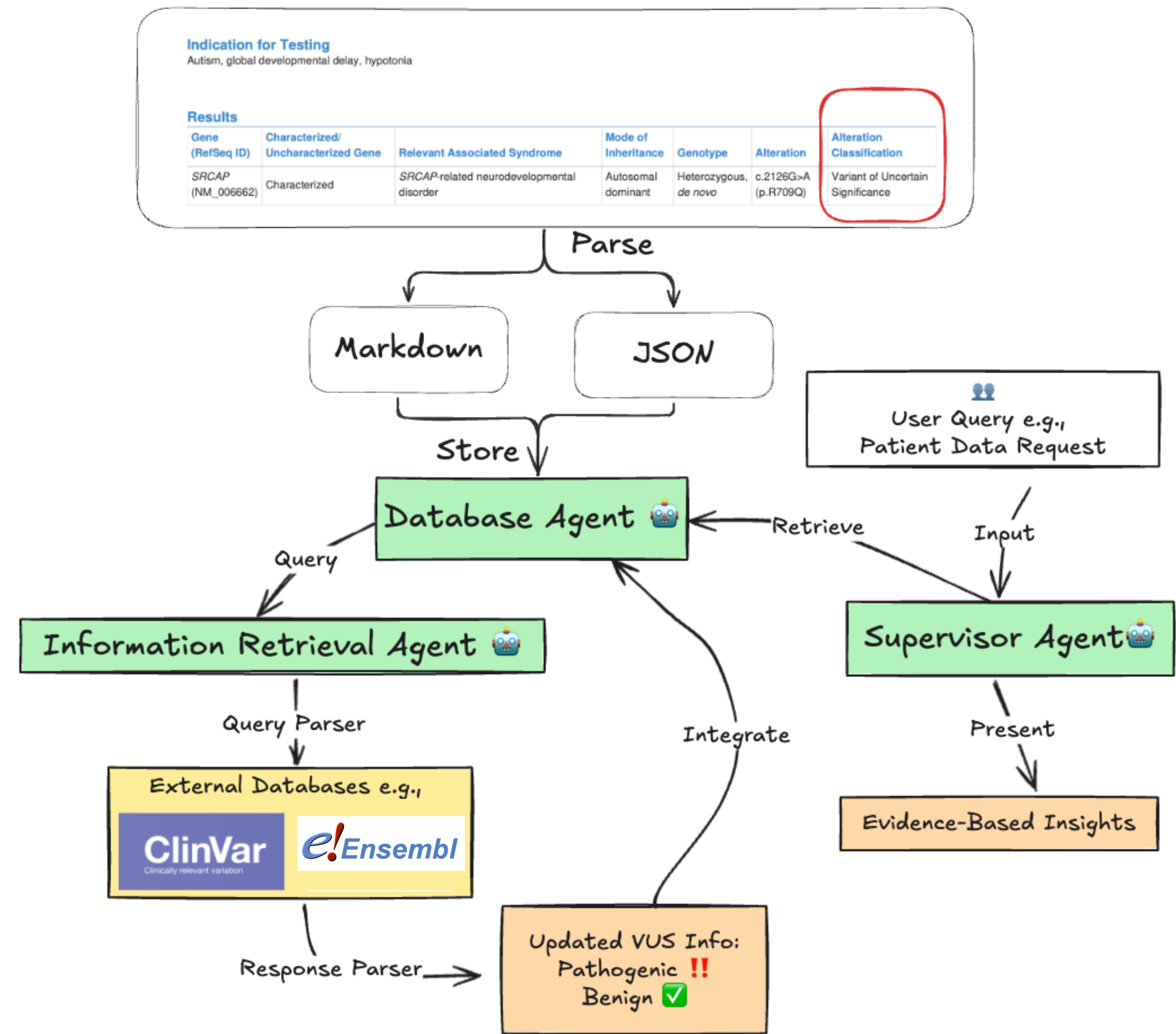
2. Variant Extraction & Validation

- Parse variant notation (gene, transcript, HGVS format)
- Validate variant syntax and genomic references

3. Multi-Source Data Enrichment

- Cross-check variant classifications using updated ClinVar and literature-based evidence
- Flag variants with conflicting or updated classifications

System Architecture (in development)



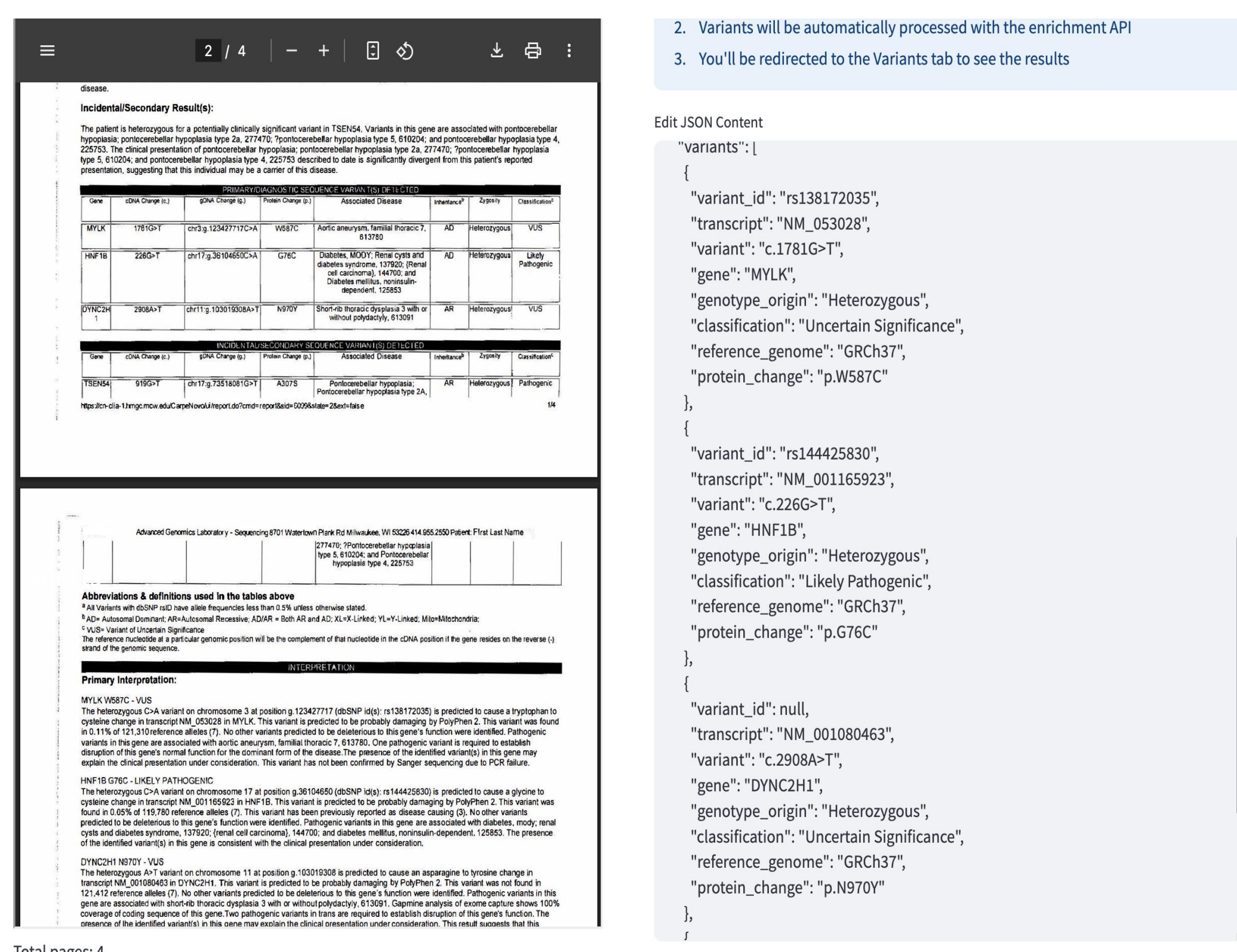
✓ OCR Parser Agent: Extracts text and structured data (e.g., gene IDs, variants) from genetic reports using Optical Character Recognition (OCR).

🗄️ Database Agent: Maintains structured genetic data in a relational database for efficient querying, caching, and updates.

✓ Information Retrieval Agent: Formats queries to external databases (e.g., ClinVar, Ensembl) for updated VUS data.

🔄 Supervisor Agent: Manages multi-agent interactions, retrieves patient-specific variant data, and provides real-time evidence-based genomic insights.

Features



The system extracts structured variant information (in JSON) from clinical reports in PDF format.

Enriched Variants

Successfully processed 3 variants

	ClinVar ID	Gene	Transcript	Variant	Classification
0	193102	HNF1B	NM_001165923.4	c.226G>T	Conflicting classifications of pathogenicity
1	1467016	DYNC2H1	NM_001080463.2	c.2908A>T	Uncertain significance
2	2120	TSEN54	NM_207346.3	c.919G>T	Pathogenic

Variant Details [Gene Information](#)

Gene Coordinate Information

This table shows genomic coordinates and reference information for the genes associated with these variants

	Ensembl ID	Gene Symbol	Position GRCh38	HGNC URL
0	ENSG00000275410	HNF1B	17:37686431-37745059	https://www.genenames.org
1	ENSG00000187240	DYNC2H1	11:103109410-103479863	https://www.genenames.org
2	ENSG00000182173	TSEN54	17:75515944-75524735	https://www.genenames.org

Download Gene Data as CSV

The **Information Retrieval Agent** queries external databases (e.g., ClinVar, Ensembl) for updated variant classifications and genomic data.

Significant Classification Findings

Found 2 variants with classifications other than 'Uncertain Significance':

	Gene	Transcript	Variant	Classification	ClinVar ID	Protein Change
0	HNF1B	NM_001165923.4	c.226G>T	Conflicting classifications of pathogenicity	193102	NP_001159395.1:p.Gly76Cys
1	TSEN54	NM_207346.3	c.919G>T	Pathogenic	2120	NP_097229.2:p.Ala307Ser

Reclassification Alerts: Flags variants with updated classifications for clinician review.

Found 1 variants that could not be validated:

Warning 1: No ClinVar IDs found

```
{  "gene": "MYLK",  "transcript": "NM_053028",  "variant": "c.1781G>T",  "variant_id": "rs138172035",  "genotype_origin": "Heterozygous",  "classification": "Uncertain Significance",  "reference_genome": "GRCh37",  "protein_change": "p.W587C",  "hgvs": "",  "hgvs_link": ""}
```

These variants need attention. Please verify the transcript, gene, and variant fields.

Missing Data Warnings: Notifies users when variant data is incomplete or unavailable.

References

- BioAgents: Democratizing Bioinformatics Analysis with Multi-Agent Systems
- ClinVar database - <https://www.ncbi.nlm.nih.gov/clinvar/>
- What is a Multiagent System? | IBM

Future Work

- Database agent for local caching of frequently accessed data
- Workflow supervisor for resource management and error recovery
- Automated retrieval of up-to-date variant reclassification data from PubMed and biomedical databases